

# Keqing Liu

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## Professional Summary

Quantitative Macro Economist and Data Scientist with extensive expertise in building predictive and structural macroeconomic models (DSGE) alongside machine learning algorithms (e.g., Random Forest). Proven ability to leverage high-dimensional datasets to simulate how agents (households, firms, and governments) forward-look and react to strategic shocks. Adept at running scalable counterfactual simulations to forecast economic trends and evaluate policy or business impacts—such as quantifying the systemic effects of interest rate hikes or geopolitical tensions. Skilled in Python and MATLAB, with a strong record of translating complex quantitative analysis into clear, strategic recommendations for both technical and non-technical audiences.

## Technical Skills

- **Quantitative Modeling:** DSGE models, statistical modeling, econometric estimation, time series analysis (e.g., ARIMA, SVAR), machine learning (e.g., Random Forest), forecasting
- **Programming:** Python (NumPy, Pandas, SciPy, Scikit-learn), SQL, MATLAB, R, Stata, Git
- **Data Analysis:** Regression analysis, data visualization, counterfactual analysis, causal inference

## Education

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|-----------|-----------------------------|---------------------------------------------|
| 2013-2017 | Ph.D. in Economics          | University of Exeter (ESRC scholarship)     |
| 2012-2013 | MSc Economics (Distinction) | University of Exeter                        |
| 2008-2012 | B.A. in Economics           | Huazhong University of Science & Technology |

## Work Experience

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|-------------------|---------------------|----------------------------------------------|
| 2025.09 - Present | Associate Professor | Xiamen University                            |
| 2018.09 - 2025.08 | Assistant Professor | Xiamen University                            |
| 2017.08 - 2018.07 | Associate Professor | Shanghai University of Finance and Economics |

## Research Expertise

I specialize in integrating macro-financial economic theory with advanced data science techniques to solve complex forecasting, risk assessment, and policy evaluation problems. My structural and empirical expertise includes:

- **Economic Modeling & Policy Simulation:** Expert in developing and estimating Dynamic Stochastic General Equilibrium (DSGE) models with frictional banking sectors. Adept at applying nonlinear and perturbation solution methods to run high-dimensional “what-if” policy experiments, simulating and forecasting the business cycle impacts of regulatory or interest rate shocks under diverse economic scenarios.

- **Machine Learning & Risk Classification:** Developed a Random Forest classification model using Chinese bank-level balance sheet data, loan portfolio metrics, capital adequacy ratios, non-performing loan indicators, liquidity measures, asset concentration, maturity mismatch, and real estate/corporate credit exposure, combined with regional macroeconomic variables. The model captured nonlinear relationships between bank operating characteristics and financial risk events, enabling early-warning predictions of significant increases in non-performing loan ratios, deterioration in regulatory risk ratings, and other indicators of financial instability. This approach linked micro-level bank fundamentals with regional macro conditions to improve credit default risk assessment and systemic risk monitoring.
- **Time-Series Estimation & Econometrics:** Proficient in advanced statistical modeling and econometric methods (e.g., ARIMA, SVAR). Developed integrated New Keynesian frameworks that combine structural micro-foundations with time-series estimation techniques to solve optimal policy and time-inconsistency problems.

## Policy Experience

I have direct experience translating research into policy insights. During my tenure at Shanghai University of Finance and Economics, I also served as a Senior Research Associate, contributing to a team that produced quarterly policy research reports on China's economy for the central bank and government institutions, providing actionable strategic guidance directly utilized by senior policymakers. My focus within the group was on monetary policy analysis.

## Selected Publications

- [1] Habit formation and news-driven business cycles (2025), *Review of Economic Dynamics*, 56, 101273, with Fengqi Liu and Jianpo Xue.
- [2] Should macroprudential policy be countercyclical? (2024), *Journal of Economic Dynamics and Control*, 158, 104765, with Yoske Igarashi.
- [3] Credit expansion, bank liberalization, and structural change in bank asset accounts (2021), *Journal of Economic Dynamics and Control*, 124, 104066, with Michael Fan.

## Selected Conference Presentations

- [1] Discussant. Economic Cycles and Growth Conference, in Xiamen, 2024: 'Uncovering Frictions to Long-Term Borrowing'.
- [2] Asian Meeting of the Econometric Society (AMES) in Xiamen, 2019: 'Credit Expansion, Bank Liberalization, and Structural Change in Bank Asset Accounts'.
- [3] Money Macro Finance Annual Conference (MMF) in Bath, 2016: 'Bank Equity and Macroprudential Policy'.

## Teaching (in English)

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|------------------------------------------------------|-----------|
| Money and Banking                                    | 2018-2025 |
| Advanced Macroeconomics: Money, Banking, and Finance | 2023-2025 |
| Advanced Macroeconomics II                           | 2019-2022 |
| Intermediate Microeconomics (SUFU)                   | 2017-2018 |